

Mohammed Mowla

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Education

Stockton University

Bachelor of Science in Computer Science, Minor in Mathematics

GPA: 3.78 | Honors: Stockton University Provost's Scholarship (4 Years)

May 2024

Galloway, NJ

Skills

Languages: Python, Java, TypeScript, JavaScript, C#, HTML/CSS, PHP

Databases & Cloud: Supabase, PostgreSQL, MySQL, AWS S3, Vercel

Other: Selenium, PyTest, Appium, Next.js, Express.js, Jakarta EE, React, JUnit, Svelte, Tailwind CSS, JavaFX, Git/GitHub, Apache Tomcat, Node.js, Three.js, p5.js, NumPy, Matplotlib, SciPy, PyTorch

Experience

Computer Scientist - Software Automation

Oct. 2024 - Present

LS Technologies: Computer Scientist – A.M.M.S.

Egg Harbor Township, NJ

- Developed a **Python**, **Selenium**, and **Appium** regression framework across a **web** and an **iOS** application, resulting in an **88%** increase in testing coverage, safeguarding core processes
- Engineered a **CI/CD** pipeline that eliminated manual setup for up to **15** daily development build releases
- Conducted automated and manual testing within an **Agile** environment by employing a deep knowledge of a suite of interconnected enterprise applications to design **black-box** test cases that aligned with real-world user scenarios
- Validated functionality with Subject Matter Experts (**SMEs**), sharing findings with development to resolve defects

Computer Scientist Intern

June 2024 - Aug. 2024

Federal Aviation Administration – A.M.M.S.

Egg Harbor Township, NJ

- Optimized automation runs by **33%** for the **iOS** regression suite by refining test scripts and simulator configurations
- Expanded manual and automated test coverage to align with Test and Evaluation (**T&E**) practices
- Directed **11** interns in the development of the Air Traffic Safety Management Team's organizational **Wiki** using **Confluence** and **Jira** integrations to improve knowledge sharing

Computer Scientist Intern

June 2023 - Aug. 2023

Federal Aviation Administration – A.M.M.S.

Egg Harbor Township, NJ

- Spearheaded the utilization of **Appium** automation to establish a regression suite for an **iOS** application in **Java**
- Deployed and benchmarked the Simulated Driver Radar Recorder (**SDRR**) and its ecosystem on a cloud-hosted **Ubuntu** instance via **MobaXterm**
- Co-led a cohort project researching virtualization of test systems, mentoring peers, and substantially accelerating lab configuration

Personal Projects

Finite Automata Designer

February 2025 - Present

- Built and deployed a full-stack web application using **Next.js** and **TypeScript** that lets users visually design, edit, and simulate deterministic (**DFA**) and nondeterministic (**NFA**) finite automata, hosted on **Vercel**
- Engineered an interactive **HTML5 Canvas** editor with drag-and-drop state creation, transition drawing, **real-time rendering**, building custom hit-detection and a self-managed scene graph
- Implemented a Theory of Computation engine to simulate state transitions and validate input strings against user-built automata, returning acceptance and rejection results
- Integrated **Supabase** authentication and persistence, serializing each automaton's state-transition graph to **JSON** and applying row-level security so users can securely save, load, and revisit their designs
- Maintained an active open-source codebase of **700+** commits using **Git** issues and pull requests